

researcher in the laboratories which made the transition from molecular genetics to genetic engineering at that time. See also T. 500/91, in which the board ruled that the development of the art normally expected by the notionally skilled person – who may be represented by a team of appropriate specialists (T. 141/87) – did not include solving technical problems by performing scientific research in areas not yet explored (see also T. 441/93).

In T. 223/92 the board had to consider the knowledge and capabilities of the notional skilled person in the field of genetic engineering as at October 1981, more than one year later than was the case in T. 500/91. By this time, a considerably greater number of genes had been made the subject of cloning and expressing methods, and skills and experience in this technical field were developing rapidly. The knowledge of the notional person skilled in the art had to be considered as that of a team of appropriate specialists who knew all the difficulties still to be expected when considering the cloning of a new gene. However, the skilled person had to be assumed to lack the inventive imagination to solve problems for which routine methods of solution did not already exist.

From the notional skilled person nothing more can be expected than the carrying out of experimental work by routine means within the framework of the normal practice of filling gaps in knowledge by the application of existing knowledge (T. 886/91, T. 223/92, T. 530/95, T. 791/96).

In T. 207/94 (OJ 1999, 273) the board stated that it had to be assumed that the average skilled person would not engage in creative thinking. Yet he or she could be expected to react in a way common to all skilled persons at any time, namely that an assumption or hypothesis about a possible obstacle to the successful realisation of a project must always be based on facts. Thus, in the board's view, an absence of evidence that a given feature might be an obstacle to carrying out an invention would not be taken as an indication that this invention could not be achieved, nor that it could.

In T. 412/93 the patent related to the production of erythropoietin. The parties agreed that in this particular case the skilled person should be treated as a team of three, composed of one PhD researcher with several years' experience in the aspect of gene technology or biochemistry under consideration, assisted by two laboratory technicians fully acquainted with the known techniques relevant to that aspect. The composition of the team might vary depending on the knowledge and skills required by the particular aspect dealt with.

In T. 455/91 (OJ 1995, 684) the board held that the skilled person in this field was well aware that even a small structural change in a product (e.g. a vector, protein, or DNA sequence) or procedure (e.g. a purification process) could produce dramatic functional changes. He would therefore adopt a conservative attitude. For example, he would neither go against an established prejudice, nor venture into "sacrosanct" or unpredictable areas, nor take incalculable risks. However, within the normal design procedures, he would readily seek appropriate, manifest changes, modifications or adjustments involving little trouble or work and no or only calculable risks, especially to obtain a handier or more convenient product or simplify a procedure. See also T. 441/93, T. 867/13.

If, on the other hand, he would expect to have to perform scientific research rather than routine work in order to transfer a technology previously set up in one field of research (method of transforming *Saccharomyces cerevisiae* whole cells) to a neighbouring field (method of transforming *Kluyveromyces* whole cells), then inventive step could be acknowledged (T 441/93).

In T 493/01 the invention related to a protective antigen potentially useful in a vaccine against whooping cough. The board noted that in T 455/91 (OJ 1995, 684) the skilled person in the field of biotechnology had already been defined as being cautious and conservative. The board said this did not mean he would refrain from considering information because it did not concern the mainstream of research in his field of specialisation or because it applied only to some parts of the world. His skill and knowledge were not geographically limited; in fact he would have a global point of view. Thus if, as in the case in point, a pathogen constituted a known threat in some restricted parts of the world, the skilled person would not refrain from taking prior knowledge about that pathogen into consideration or from using it as a basis for his activities.

8.1.4 Identification of the skilled person in the case of computer-implemented inventions

In T 641/00 (OJ 2003, 352) the board stated that the identification of the skilled person needed careful consideration. He would be an **expert in a technical field**. If the technical problem concerned a computer implementation of a business, actuarial or accountancy system, he would be someone skilled in data processing, and not merely a businessman, actuary or accountant (T 172/03).

In T 531/03 the patentee submitted that the invention at issue required a combination of a technical and a non-technical inventive step, and that the skilled persons would therefore consist of a team of a "non-technical person" plus a technical person. The board rejected this approach and stated that an attempt to take into account the contribution of non-technical and technical aspects on an equal footing in the assessment of inventive step would be inconsistent with the EPC, since the presence of inventive step would, in such an approach, be attributed to features which were defined in the EPC as not being an invention.

The board in T 407/11 held that the relevant skilled person in the context of providing computer-system users with operating assistance via a user interface (e.g. error messages or warnings) was an expert in software ergonomics concerned with the user-friendliness of human-machine interfaces rather than an expert in software programming or in computer technology in the strict sense.

8.2. Neighbouring field

Two landmark decisions, T 176/84 (OJ 1986, 50) and T 195/84 (OJ 1986, 121) addressed in detail the problem of the relevant technical field, i.e. the question of the extent to which neighbouring areas beyond the specific field of the application might be taken into consideration when assessing inventive step. According to T 176/84 the state of the art to be considered when examining for inventive step includes, as well as that in the specific

field of the application, the state of any relevant art in neighbouring fields and/or a broader general field of which the specific field is part, that is to say any field in which the same problem or one similar to it arises and of which the person skilled in the art of the specific field must be expected to be aware. The board in T 195/84 added that the state of the art to be considered included that in a non-specific (general) field dealing with the solution of any general technical problem which the application sought to solve in its specific field. Such solutions of general technical problems in non-specific (general) fields had to be considered as forming part of the general technical knowledge which a priori was to be attributed to those skilled persons versed in any specific technical field. These principles have been applied in a large number of decisions.

In T 560/89 (OJ 1992, 725) the board took the view that the skilled person would also draw on prior art in other fields which were neither neighbouring nor broader general fields, if prompted to do so because the materials used were related or because of public debate about a technical problem common to both fields. Expanding on this, T 955/90 added that, in practice, the person skilled in a broader general field would also draw on the narrower, more specialised field of the known main application of the general technology in search of a solution to a problem lying outside the special application of that technology (T 379/96).

In T 454/87 the board expressed the opinion that a skilled person specialising in a particular technical field (gas chromatography equipment) would, in the course of his normal professional activity, also observe developments in equipment used in a related technical field (absorption spectral analysis).

In T 891/91 the board stated that a skilled person in the field of lenses for ophthalmic use, confronted with the technical problem of adhesion and abrasion resistance of a coating made on a surface of the lens, would also refer to the state of the art in the more general field of coated plastic sheets in which the same problems of adhesion and abrasion resistance of the coating arose and of which he was aware.

In T 1910/11 the board stated that, whether the prior art and the claimed invention belonged to neighbouring technical fields within the meaning of T 176/84 was less a question of whether the relevant implementation parameters were identical than a question of the degree of similarity between the respective problems, boundary conditions and functional concepts. Applying this to the case in hand, it observed that automotive electronics and avionics were traditionally considered to be neighbouring technical fields because they involved similar problems (e.g. interference resistance, robustness and reliability), boundary conditions (e.g. mobility) and functional concepts (e.g. physical/logical separation of the communications systems for safety and maintenance data in the vehicle).

In T 767/89, regarding carpets, the board ruled that wigs were neither a neighbouring technical field nor a broader general one which included the former field. So wigs were not a related technical field in which the person skilled in carpets would have been prompted to seek solutions. The two inventions addressed different problems; the user requirements were not comparable.

In T.675/92 the board held that, because of the differing security risks, a skilled person could not be expected to search in the field of bulk-goods packaging for ideas for the design of a closure for a means of conveying money.

Further comments on the concept of relevant field are to be found in several other decisions, including the following: T.277/90 (in dentistry, moulding technology and prosthodontics are neighbouring technical fields), T.358/90 (discharging the content of a portable toilet did not lead the skilled person to the field of filling a tank of a chain saw by means of a special kind of container), T.1037/92 (a person skilled in the art of making fuse links for programmable ROMs would also have consulted the documentation in the field of ultraminiaturised integrated switches), T.838/95 (the pharmaceutical and cosmetic fields were immediate neighbours), T.26/98 (the board did not consider the field of electrochemical generators to be a neighbouring field of iontophoresis because, though both fields relied on electrochemical processes, such processes had substantially different purposes and applications and, consequently, had to satisfy different requirements), T.1202/02 (the manufacture of mineral fibres and that of glass fibres were two closely related technical fields irrespective of differences between the raw materials respectively used), T.47/91 (casting rolls and casting wheels when continuously casting molten metal represent neighbouring technical fields).

In T.244/91 the board found the person skilled in the art of heat exchangers would take account of publication D3 on "gaskets of plate heat exchangers", or would at least consult a specialist in the non-specific field of gaskets who must be expected to be familiar with such standard literature .

In T.1154/16 the board found that the disclosures of D9 and D19 in the case in hand could not be considered to be in a neighbouring field to that of shoe press belts for papermaking, but that they did form part of a broader, general field, relating to the general use of polyurethanes in dynamic mechanical applications solving similar problems to the current case. Therefore the skilled person would consider the disclosure of both documents.

In T.1979/18 the board found that while it was true that agricultural engineering may combine (depending on the particular application) with a wide variety of other disciplines such as civil, chemical, mechanical, bio-systems, and environmental engineering, this did not mean that the skilled person was therefore also fully knowledgeable in those fields.

In T.28/87 (OJ 1989, 383) the board held that if reference was made in the introduction to the description of an application or a patent to a state of the art which could not objectively be classified as a relevant field, that state of the art could not in the course of examination for patentability be applied to the applicant's or patent proprietor's disadvantage as a neighbouring field merely on account of that reference.

8.3. Skilled person – level of knowledge

Inventive step has to be assessed on the basis of the skilled person's knowledge before the priority or filing date (see (T.268/89, OJ 1994, 50; T.365/89).

The same level of skill has to be applied when, for the same invention, the two questions of sufficient disclosure and inventive step have to be considered (T. 60/89, OJ 1992, 268; T. 373/94, T. 99/14, T. 1861/16, see also chapter II.C.4.1 "The disclosure is aimed at the skilled person"). In T. 694/92 (OJ 1997, 408) the board added that, although the same level of skill is applied for both Art. 56 and Art. 83 EPC 1973, the two starting points differ; for inventive step purposes, the skilled person knows only the prior art; for sufficiency of disclosure, he knows the prior art **and** the disclosed invention.

In T. 426/88 (OJ 1992, 427) the board held that a book providing general teaching in a general technical field covering the invention's specific technical field was part of the general knowledge of a specialist in that specific technical field. When books, representing common general knowledge, described a basic general technical theory or methodology and exemplified the same with specific applications in certain technical fields only, these did not limit the general scope and relevance of such disclosures so as to exclude possible applications in other fields. The appellant had argued that the book, written in German, was not a general reference book consulted by experts in that field in Great Britain. The board, however, adhered to the definition of the state of the art given in Art. 54 EPC 1973, according to which no account was taken of the location at which the skilled person exercised his profession. It was also of the view that the language of publication alone could not be decisive for the admissibility of a technical book representing the common general knowledge of the skilled person (see also T. 1688/08). In T. 2058/18 the board found that the (technically) skilled person might be considered a multilingual person but not normally a linguist.

In T. 632/10 the board pointed out that even if an invention happened to be obvious only for skilled persons of German nationality or residence, it would still lack an inventive step within the meaning of Art. 56 EPC 1973. The fact that the German Signature Law (SigV) was valid only within Germany thus had no bearing on its status as prior art or its relevance for the assessment of inventive step outside Germany.

In T. 766/91 the board pointed out that information did not usually become common knowledge through publication in a given handbook or textbook; rather it appeared in handbooks or textbooks because it was already common knowledge. For this reason, publication in an encyclopaedia, say, could normally be taken as proof that the information was not only known but was common general knowledge. In T. 378/93 the board added that the same applied to articles in scientific periodicals addressed primarily to qualified professionals and enjoying worldwide repute. In T. 537/90 the board held that numerous publications in the specialist press over a fairly short time, reporting on meetings and research in a particularly active field of technology, could reflect common general knowledge in this field at that time.

In T. 939/92 (OJ 1996, 309) it was explained that the state of the art could also perfectly well reside solely in the relevant common general knowledge, which, in turn, need not necessarily be in writing, i.e. in textbooks or the like, but might simply be a part of the unwritten "mental furniture" of the average skilled person. In the case of any dispute, however, the extent of the relevant common general knowledge had to be proven, e.g. by documentary or oral evidence.